

Thermodynamics

Instructor

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Chapter 16 – Problems

**Problem numbers correspond to
5th Edition**

K_p and the Equilibrium Composition of Ideal Gases

16-14 Determine the equilibrium constant K_p for the process $\text{CO} + \frac{1}{2}\text{O}_2 = \text{CO}_2$ at (a) 298 K and (b) 2000 K. Compare your results with the values for K_p listed in Table A-28.

Analysis (a) The K_p value of a reaction at a specified temperature can be determined from the Gibbs function data using

$$K_p = e^{-\Delta G^*(T)/R_u T} \quad \text{or} \quad \ln K_p = -\Delta G^*(T) / R_u T$$

where

$$\Delta G^*(T) = \nu_{\text{CO}_2} \bar{g}_{\text{CO}_2}^*(T) - \nu_{\text{CO}} \bar{g}_{\text{CO}}^*(T) - \nu_{\text{O}_2} \bar{g}_{\text{O}_2}^*(T)$$

At 298 K,

$$\Delta G^*(T) = 1(-394,360) - 1(-137,150) - 0.5(0) = -257,210 \text{ kJ/kmol}$$

where the Gibbs functions are obtained from Table A-26. Substituting,

$$\ln K_p = -\frac{(-257,210 \text{ kJ/kmol})}{(8.314 \text{ kJ/kmol} \cdot \text{K})(298 \text{ K})} = \mathbf{103.81}$$

From Table A-28: $\ln K_p = \mathbf{103.76}$

(b) At 2000 K,

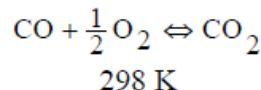
$$\begin{aligned} \Delta G^*(T) &= \nu_{\text{CO}_2} \bar{g}_{\text{CO}_2}^*(T) - \nu_{\text{CO}} \bar{g}_{\text{CO}}^*(T) - \nu_{\text{O}_2} \bar{g}_{\text{O}_2}^*(T) \\ &= \nu_{\text{CO}_2} (\bar{h} - T\bar{s})_{\text{CO}_2} - \nu_{\text{CO}} (\bar{h} - T\bar{s})_{\text{CO}} - \nu_{\text{O}_2} (\bar{h} - T\bar{s})_{\text{O}_2} \\ &= 1[(-302,128) - (2000)(309.00)] - 1[(-53,826) - (2000)(258.48)] - 0.5[(59,193) - (2000)(268.53)] \\ &= -110,409 \text{ kJ/kmol} \end{aligned}$$

The enthalpies at 2000 K and entropies at 2000 K and 101.3 kPa (1 atm) are obtained from EES. Substituting,

$$\ln K_p = -\frac{(-110,409 \text{ kJ/kmol})}{(8.314 \text{ kJ/kmol} \cdot \text{K})(2000 \text{ K})} = \mathbf{6.64}$$

From Table A-28:

$$\ln K_p = \mathbf{6.635}$$



16–16 Determine the equilibrium constant K_p for the reaction $\text{CH}_4 + 2\text{O}_2 \rightleftharpoons \text{CO}_2 + 2\text{H}_2\text{O}$ at 25°C .

Answer: 1.96×10^{140}

16–17 Using the Gibbs function data, determine the equilibrium constant K_p for the dissociation process $\text{CO}_2 \rightleftharpoons \text{CO} + \frac{1}{2}\text{O}_2$ at (a) 298 K and (b) 1800 K. Compare your results with the K_p values listed in Table A–28.

16-21

Carbon monoxide is burned with 100 percent excess air during a steady-flow process at a pressure of 1 atm. At what temperature will 97 percent of CO burn to CO_2 ? Assume the equilibrium mixture consists of CO_2 , CO, O_2 , and N_2 . *Answer: 2276 K*

Assumptions 1 The equilibrium composition consists of CO_2 , CO, O_2 , and N_2 . 2 The constituents of the mixture are ideal gases.

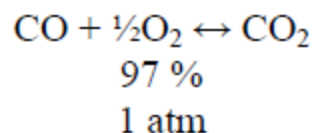
Analysis Assuming N_2 to remain as an inert gas, the stoichiometric and actual reactions can be written as

Stoichiometric: $\text{CO} + \frac{1}{2} \text{O}_2 \rightleftharpoons \text{CO}_2$ (thus $\nu_{\text{CO}_2} = 1$, $\nu_{\text{CO}} = 1$, and $\nu_{\text{O}_2} = \frac{1}{2}$)

Actual: $\text{CO} + 1(\text{O}_2 + 3.76 \text{N}_2) \longrightarrow \underbrace{0.97 \text{CO}_2}_{\text{product}} + \underbrace{0.03 \text{CO} + 0.515 \text{O}_2}_{\text{reactants}} + \underbrace{3.76 \text{N}_2}_{\text{inert}}$

The equilibrium constant K_p can be determined from

$$\begin{aligned}
 K_p &= \frac{N_{\text{CO}_2}^{\nu_{\text{CO}_2}}}{N_{\text{CO}}^{\nu_{\text{CO}}} N_{\text{O}_2}^{\nu_{\text{O}_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{\text{CO}_2} - \nu_{\text{CO}} - \nu_{\text{O}_2})} \\
 &= \frac{0.97}{0.03 \times 0.515^{0.5}} \left(\frac{1}{0.97 + 0.03 + 0.515 + 3.76} \right)^{1-1.5} \\
 &= 103.48
 \end{aligned}$$

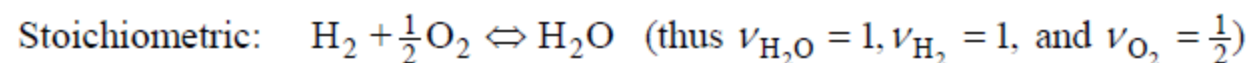


From Table A-28, the temperature corresponding to this K_p value is $T = 2276 \text{ K}$

16–24 Hydrogen is burned with 150 percent theoretical air during a steady-flow process at a pressure of 1 atm. At what temperature will 98 percent of H_2 burn to H_2O ? Assume the equilibrium mixture consists of H_2O , H_2 , O_2 , and N_2 .

Assumptions 1 The equilibrium composition consists of H_2O , H_2 , O_2 , and N_2 . 2 The constituents of the mixture are ideal gases.

Analysis Assuming N_2 to remain as an inert gas, the stoichiometric and actual reactions can be written as

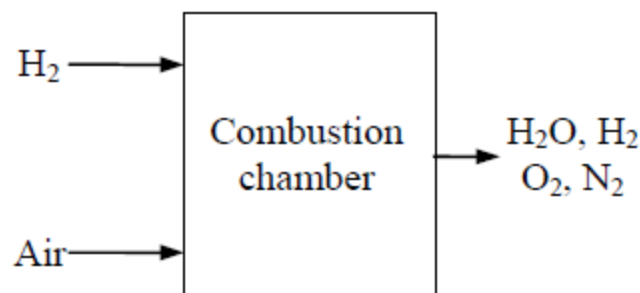


The equilibrium constant K_p can be determined from

$$K_p = \frac{N_{H_2O}^{\nu_{H_2O}}}{N_{H_2}^{\nu_{H_2}} N_{O_2}^{\nu_{O_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{H_2O} - \nu_{H_2} - \nu_{O_2})}$$

$$= \frac{0.98}{0.02 \times 0.26^{0.5}} \left(\frac{1}{0.98 + 0.02 + 0.26 + 2.82} \right)^{1-1.5}$$

$$= 194.11$$



From Table A-28, the temperature corresponding to this K_p value is $T = 2472 \text{ K}$.

16–25 Air (79 percent N_2 and 21 percent O_2) is heated to 2000 K at a constant pressure of 2 atm. Assuming the equilibrium mixture consists of N_2 , O_2 , and NO, determine the equilibrium composition at this state. Is it realistic to assume that no monatomic oxygen or nitrogen will be present in the equilibrium mixture? Will the equilibrium composition change if the pressure is doubled at constant temperature?

Assumptions 1 The equilibrium composition consists of N_2 , O_2 , and NO. 2 The constituents of the mixture are ideal gases.

Analysis The stoichiometric and actual reactions in this case are

Stoichiometric: $\frac{1}{2} N_2 + \frac{1}{2} O_2 \rightleftharpoons NO$ (thus $\nu_{NO} = 1$, $\nu_{N_2} = \frac{1}{2}$, and $\nu_{O_2} = \frac{1}{2}$)

Actual: $3.76 N_2 + O_2 \longrightarrow \underbrace{x NO}_{\text{prod.}} + \underbrace{y N_2 + z O_2}_{\text{reactants}}$

N balance: $7.52 = x + 2y$ or $y = 3.76 - 0.5x$

O balance: $2 = x + 2z$ or $z = 1 - 0.5x$

Total number of moles: $N_{\text{total}} = x + y + z = x + 4.76 - x = 4.76$

The equilibrium constant relation can be expressed as

$$K_p = \frac{N_{NO}^{\nu_{NO}}}{N_{N_2}^{\nu_{N_2}} N_{O_2}^{\nu_{O_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{NO} - \nu_{N_2} - \nu_{O_2})}$$

AIR
2000 K
2 atm



From Table A-28, $\ln K_p = -3.931$ at 2000 K. Thus $K_p = 0.01962$. Substituting,

$$0.01962 = \frac{x}{(3.76 - 0.5x)^{0.5} (1 - 0.5x)^{0.5}} \left(\frac{2}{4.76} \right)^{1-1}$$

Solving for x ,

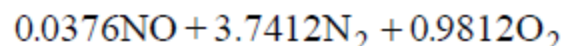
$$x = 0.0376$$

Then,

$$y = 3.76 - 0.5x = 3.7412$$

$$z = 1 - 0.5x = 0.9812$$

Therefore, the equilibrium composition of the mixture at 2000 K and 2 atm is



The equilibrium constant for the reactions $\text{O}_2 \rightleftharpoons 2\text{O}$ ($\ln K_p = -14.622$) and $\text{N}_2 \rightleftharpoons 2\text{N}$ ($\ln K_p = -41.645$) are much smaller than that of the specified reaction ($\ln K_p = -3.931$). Therefore, it is realistic to assume that no monatomic oxygen or nitrogen will be present in the equilibrium mixture. Also the equilibrium composition is in this case is independent of pressure since $\Delta \nu = 1 - 0.5 - 0.5 = 0$.

16-26 Hydrogen (H_2) is heated to 3200 K at a constant pressure of 8 atm. Determine the percentage of H_2 that will dissociate into H during this process. *Answer: 5.0 percent*

Assumptions 1 The equilibrium composition consists of H_2 and H. 2 The constituents of the mixture are ideal gases.

Analysis The stoichiometric and actual reactions can be written as

Stoichiometric: $H_2 \Leftrightarrow 2H$ (thus $\nu_{H_2} = 1$ and $\nu_H = 2$)

Actual: $H_2 \longrightarrow \underbrace{xH_2}_{\text{react.}} + \underbrace{yH}_{\text{prod.}}$

H balance: $2 = 2x + y$ or $y = 2 - 2x$

Total number of moles: $N_{\text{total}} = x + y = x + 2 - 2x = 2 - x$

The equilibrium constant relation can be expressed as

$$K_p = \frac{N_H^{\nu_H}}{N_{H_2}^{\nu_{H_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{\nu_H - \nu_{H_2}}$$

From Table A-28, $\ln K_p = -2.534$ at 3200 K. Thus $K_p = 0.07934$. Substituting,

$$0.07934 = \frac{(2 - 2x)^2}{x} \left(\frac{8}{2 - x} \right)^{2-1}$$

Solving for x , $x = 0.95$

Thus the percentage of H_2 which dissociates to H at 3200 K and 8 atm is

$$1 - 0.95 = 0.05 \text{ or } 5.0\%$$

H_2
3200 K
8 atm

16–30 A mixture of 3 mol of N_2 , 1 mol of O_2 , and 0.1 mol of Ar is heated to 2400 K at a constant pressure of 10 atm. Assuming the equilibrium mixture consists of N_2 , O_2 , Ar, and NO, determine the equilibrium composition.

Answers: 0.0823NO, 2.9589 N_2 , 0.9589 O_2 , 0.1Ar

Assumptions 1 The equilibrium composition consists of N_2 , O_2 , Ar, and NO. 2 The constituents of the mixture are ideal gases.

Analysis The stoichiometric and actual reactions in this case are

Stoichiometric: $\frac{1}{2} N_2 + \frac{1}{2} O_2 \rightleftharpoons NO$ (thus $\nu_{NO} = 1$, $\nu_{N_2} = \frac{1}{2}$, and $\nu_{O_2} = \frac{1}{2}$)

Actual: $3 N_2 + O_2 + 0.1 Ar \longrightarrow \underbrace{x NO}_{\text{prod.}} + \underbrace{y N_2 + z O_2}_{\text{reactants}} + \underbrace{0.1 Ar}_{\text{inert}}$

N balance: $6 = x + 2y \longrightarrow y = 3 - 0.5x$

O balance: $2 = x + 2z \longrightarrow z = 1 - 0.5x$

Total number of moles: $N_{\text{total}} = x + y + z + 0.1 = 4.1$

The equilibrium constant relation becomes,

$$K_p = \frac{N_{NO}^{\nu_{NO}}}{N_{N_2}^{\nu_{N_2}} N_{O_2}^{\nu_{O_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{NO} - \nu_{N_2} - \nu_{O_2})} = \frac{x}{y^{0.5} z^{0.5}} \left(\frac{P}{N_{\text{total}}} \right)^{1-0.5-0.5}$$

From Table A-28, $\ln K_p = -3.019$ at 2400 K. Thus $K_p = 0.04885$. Substituting,

$$0.04885 = \frac{x}{(3 - 0.5x)^{0.5} (1 - 0.5x)^{0.5}} \times 1$$

Solving for x ,

$$x = 0.0823$$

Then,

$$y = 3 - 0.5x = 2.9589$$

$$z = 1 - 0.5x = 0.9589$$

Therefore, the equilibrium composition of the mixture at 2400 K and 10 atm is

$$0.0823NO + 2.9589N_2 + 0.9589O_2 + 0.1Ar$$

3 N_2
1 O_2
0.1 Ar
2400 K
10 atm

16-31 Determine the mole fraction of sodium that ionizes according to the reaction $\text{Na} \rightleftharpoons \text{Na}^+ + e^-$ at 2000 K and 0.8 atm ($K_p = 0.668$ for this reaction). *Answer: 67.5 percent*

Assumptions All components behave as ideal gases.

Analysis The stoichiometric and actual reactions can be written as

Stoichiometric: $\text{Na} \rightleftharpoons \text{Na}^+ + e^-$ (thus $\nu_{\text{Na}} = 1$, $\nu_{\text{Na}^+} = 1$ and $\nu_{e^-} = 1$)

Actual: $\text{Na} \longrightarrow \underbrace{x\text{Na}}_{\text{react.}} + \underbrace{y\text{Na}^+ + ye^-}_{\text{products}}$

Na balance: $1 = x + y$ or $y = 1 - x$

Total number of moles: $N_{\text{total}} = x + 2y = 2 - x$

The equilibrium constant relation becomes,

$$K_p = \frac{N_{\text{Na}}^{\nu_{\text{Na}}} N_{e^-}^{\nu_{e^-}}}{N_{\text{Na}^+}^{\nu_{\text{Na}^+}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{\text{Na}^+} + \nu_{e^-} - \nu_{\text{Na}})} = \frac{y^2}{x} \left(\frac{P}{N_{\text{total}}} \right)^{1+1-1}$$

Substituting,

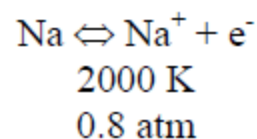
$$0.668 = \frac{(1-x)^2}{x} \left(\frac{0.8}{2-x} \right)$$

Solving for x ,

$$x = 0.325$$

Thus the fraction of Na which dissociates into Na^+ and e^- is

$$1 - 0.325 = 0.675 \quad \text{or} \quad \mathbf{67.5\%}$$

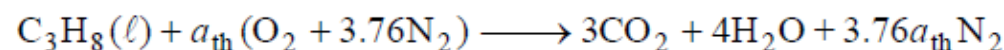


16–32 Liquid propane (C_3H_8) enters a combustion chamber at 25°C at a rate of 1.2 kg/min where it is mixed and burned with 150 percent excess air that enters the combustion chamber at 12°C . If the combustion gases consist of CO_2 , H_2O , CO , O_2 , and N_2 that exit at 1200 K and 2 atm , determine (a) the equilibrium composition of the product gases and (b) the rate of heat transfer from the combustion chamber. Is it realistic to disregard the presence of NO in the product gases?

Answers: (a) 3CO_2 , 7.5O_2 , $4\text{H}_2\text{O}$, 47N_2 , (b) 5066 kJ/min

Assumptions 1 The equilibrium composition consists of CO_2 , H_2O , CO , N_2 , and O_2 . 2 The constituents of the mixture are ideal gases.

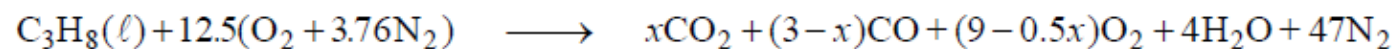
Analysis (a) Considering 1 kmol of C_3H_8 , the stoichiometric combustion equation can be written as



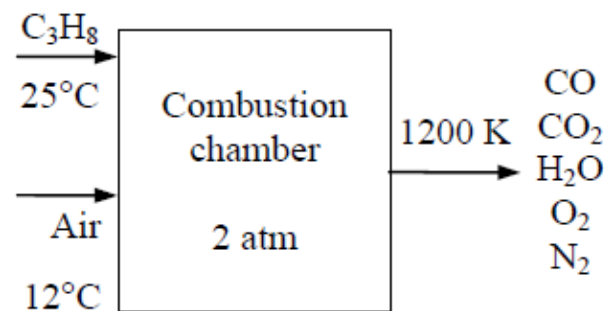
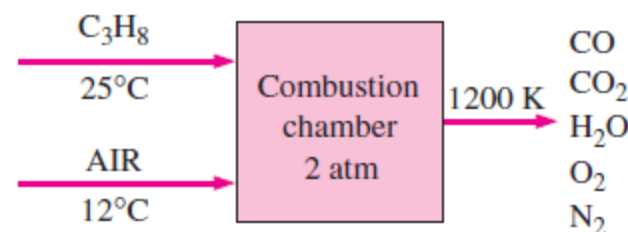
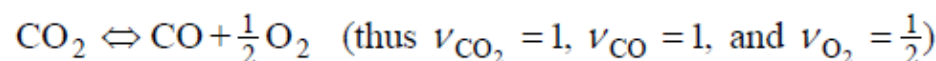
where a_{th} is the stoichiometric coefficient and is determined from the O_2 balance,

$$2.5a_{\text{th}} = 3 + 2 + 1.5a_{\text{th}} \longrightarrow a_{\text{th}} = 5$$

Then the actual combustion equation with 150% excess air and some CO in the products can be written as



After combustion, there will be no C_3H_8 present in the combustion chamber, and H_2O will act like an inert gas. The equilibrium equation among CO_2 , CO , and O_2 can be expressed as



and

$$K_p = \frac{N_{\text{CO}}^{\nu_{\text{CO}}} N_{\text{O}_2}^{\nu_{\text{O}_2}}}{N_{\text{CO}_2}^{\nu_{\text{CO}_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{\text{CO}} + \nu_{\text{O}_2} - \nu_{\text{CO}_2})}$$

where

$$N_{\text{total}} = x + (3 - x) + (9 - 0.5x) + 4 + 47 = 63 - 0.5x$$

From Table A-28, $\ln K_p = -17.871$ at 1200 K. Thus $K_p = 1.73 \times 10^{-8}$. Substituting,

$$1.73 \times 10^{-8} = \frac{(3 - x)(9 - 0.5x)^{0.5}}{x} \left(\frac{2}{63 - 0.5x} \right)^{1.5-1}$$

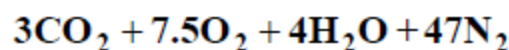
Solving for x ,

$$x = 2.99999999 \cong 3.0$$

Therefore, the amount CO in the product gases is negligible, and it can be disregarded with no loss in accuracy. Then the combustion equation and the equilibrium composition can be expressed as



and



(b) The heat transfer for this combustion process is determined from the steady-flow energy balance $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$ on the combustion chamber with $W = 0$,

$$-Q_{\text{out}} = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R$$



Assuming the air and the combustion products to be ideal gases, we have $h = h(T)$. From the tables, (The $\bar{h}_f^\circ$ of liquid propane is obtained by adding the h_{fg} at 25°C to $\bar{h}_f^\circ$ of gaseous propane).

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{285\text{ K}}$ kJ/kmol	$\bar{h}_{298\text{ K}}$ kJ/kmol	$\bar{h}_{1200\text{ K}}$ kJ/kmol
$\text{C}_3\text{H}_8 (\ell)$	-118,910	---	---	---
O_2	0	8696.5	8682	38,447
N_2	0	8286.5	8669	36,777
$\text{H}_2\text{O} (g)$	-241,820	---	9904	44,380
CO_2	-393,520	---	9364	53,848

Substituting,

$$\begin{aligned}
 -Q_{\text{out}} &= 3(-393,520 + 53,848 - 9364) + 4(-241,820 + 44,380 - 9904) \\
 &\quad + 7.5(0 + 38,447 - 8682) + 47(0 + 36,777 - 8669) \\
 &\quad - 1(-118,910 + h_{298} - h_{285}) - 12.5(0 + 8296.5 - 8682) \\
 &\quad - 47(0 + 8186.5 - 8669) \\
 &= -185,764 \text{ kJ / kmol of } \text{C}_3\text{H}_8
 \end{aligned}$$

or

$$Q_{\text{out}} = 185,764 \text{ kJ / kmol of } \text{C}_3\text{H}_8$$

The mass flow rate of C_3H_8 can be expressed in terms of the mole numbers as

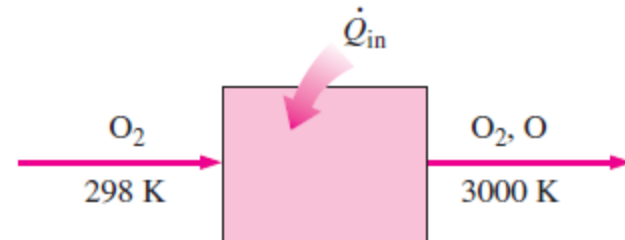
$$\dot{N} = \frac{\dot{m}}{M} = \frac{1.2 \text{ kg / min}}{44 \text{ kg / kmol}} = 0.02727 \text{ kmol / min}$$

Thus the rate of heat transfer is

$$\dot{Q}_{\text{out}} = \dot{N} \times Q_{\text{out}} = (0.02727 \text{ kmol/min})(185,746 \text{ kJ/kmol}) = \mathbf{5066 \text{ kJ/min}}$$

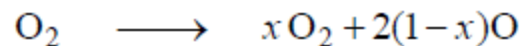
The equilibrium constant for the reaction $\frac{1}{2} \text{N}_2 + \frac{1}{2} \text{O}_2 \rightleftharpoons \text{NO}$ is $\ln K_p = -7.569$, which is very small. This indicates that the amount of NO formed during this process will be very small, and can be disregarded.

16-35 Oxygen (O_2) is heated during a steady-flow process at 1 atm from 298 to 3000 K at a rate of 0.5 kg/min. Determine the rate of heat supply needed during this process, assuming (a) some O_2 dissociates into O and (b) no dissociation takes place.

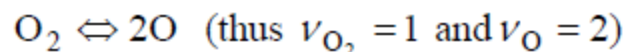


Assumptions 1 The equilibrium composition consists of O_2 and O. 2 All components behave as ideal gases.

Analysis (a) Assuming some O_2 dissociates into O, the dissociation equation can be written as



The equilibrium equation among O_2 and O can be expressed as



Assuming ideal gas behavior for all components, the equilibrium constant relation can be expressed as

$$K_p = \frac{N_O^{\nu_O}}{N_{O_2}^{\nu_{O_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{\nu_O - \nu_{O_2}}$$

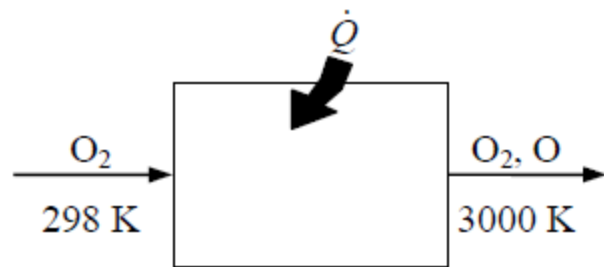
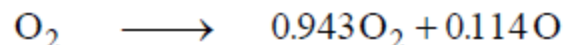
where $N_{\text{total}} = x + 2(1-x) = 2-x$

From Table A-28, $\ln K_p = -4.357$ at 3000 K. Thus $K_p = 0.01282$. Substituting,

$$0.01282 = \frac{(2-2x)^2}{x} \left(\frac{1}{2-x} \right)^{2-1}$$

Solving for x gives $x = 0.943$

Then the dissociation equation becomes



The heat transfer for this combustion process is determined from the steady-flow energy balance $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$ on the combustion chamber with $W = 0$,

$$Q_{\text{in}} = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R$$

Assuming the O_2 and O to be ideal gases, we have $h = h(T)$. From the tables,

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{298 \text{ K}}$ kJ/kmol	$\bar{h}_{3000 \text{ K}}$ kJ/kmol
O	249,190	6852	63,425
O_2	0	8682	106,780

Substituting,

$$Q_{\text{in}} = 0.943(0 + 106,780 - 8682) + 0.114(249,190 + 63,425 - 6852) - 0 = 127,363 \text{ kJ / kmol } \text{O}_2$$

The mass flow rate of O_2 can be expressed in terms of the mole numbers as

$$\dot{N} = \frac{\dot{m}}{M} = \frac{0.5 \text{ kg/min}}{32 \text{ kg/kmol}} = 0.01563 \text{ kmol/min}$$

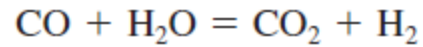
Thus the rate of heat transfer is

$$\dot{Q}_{\text{in}} = \dot{N} \times Q_{\text{in}} = (0.01563 \text{ kmol/min})(127,363 \text{ kJ/kmol}) = \mathbf{1990 \text{ kJ/min}}$$

(b) If no O_2 dissociates into O , then the process involves no chemical reactions and the heat transfer can be determined from the steady-flow energy balance for nonreacting systems to be

$$\dot{Q}_{\text{in}} = \dot{m}(h_2 - h_1) = \dot{N}(\bar{h}_2 - \bar{h}_1) = (0.01563 \text{ kmol/min})(106,780 - 8682) \text{ kJ/kmol} = \mathbf{1533 \text{ kJ/min}}$$

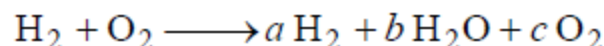
16-36 Estimate K_p for the following equilibrium reaction at 2500 K:



At 2000 K it is known that the enthalpy of reaction is -26176 kJ/kmol and K_p is 0.2209. Compare your result with the value obtained from the definition of the equilibrium constant.

16-37 A constant-volume tank contains a mixture of 1 kmol H_2 and 1 kmol O_2 at 25°C and 1 atm. The contents are ignited. Determine the final temperature and pressure in the tank when the combustion gases are H_2O , H_2 , and O_2 .

Analysis The reaction equation with products in equilibrium is

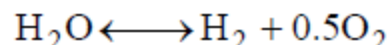


The coefficients are determined from the mass balances

$$\text{Hydrogen balance:} \quad 2 = 2a + 2b$$

$$\text{Oxygen balance:} \quad 2 = b + 2c$$

The assumed equilibrium reaction is



The K_p value of a reaction at a specified temperature can be determined from the Gibbs function data using

$$K_p = e^{-\Delta G^*(T)/R_u T} \quad \text{or} \quad \ln K_p = -\Delta G^*(T) / R_u T$$

where

$$\Delta G^*(T) = \nu_{\text{H}_2} \bar{g}_{\text{H}_2}^*(T_{\text{prod}}) + \nu_{\text{O}_2} \bar{g}_{\text{O}_2}^*(T_{\text{prod}}) - \nu_{\text{H}_2\text{O}} \bar{g}_{\text{H}_2\text{O}}^*(T_{\text{prod}})$$

and the Gibbs functions are given by

$$\bar{g}_{\text{H}_2}^*(T_{\text{prod}}) = (\bar{h} - T_{\text{prod}} \bar{s})_{\text{H}_2}$$

$$\bar{g}_{\text{O}_2}^*(T_{\text{prod}}) = (\bar{h} - T_{\text{prod}} \bar{s})_{\text{O}_2}$$

$$\bar{g}_{\text{H}_2\text{O}}^*(T_{\text{prod}}) = (\bar{h} - T_{\text{prod}} \bar{s})_{\text{H}_2\text{O}}$$



The equilibrium constant is also given by

$$K_p = \frac{a^1 c^{0.5}}{b^1} \left(\frac{P}{N_{\text{tot}}} \right)^{1+0.5-1} = \frac{ac^{0.5}}{b} \left(\frac{P_2 / 101.3}{a+b+c} \right)^{0.5}$$

An energy balance on the tank under adiabatic conditions gives

$$U_R = U_P$$

where

$$\begin{aligned} U_R &= 1(\bar{h}_{\text{H}_2@25^\circ\text{C}} - R_u T_{\text{reac}}) + 1(\bar{h}_{\text{O}_2@25^\circ\text{C}} - R_u T_{\text{reac}}) \\ &= 0 - (8.314 \text{ kJ/kmol.K})(298.15 \text{ K}) + 0 - (8.314 \text{ kJ/kmol.K})(298.15 \text{ K}) = -4958 \text{ kJ/kmol} \\ U_P &= a(\bar{h}_{\text{H}_2@T_{\text{prod}}} - R_u T_{\text{prod}}) + b(\bar{h}_{\text{H}_2\text{O}@T_{\text{prod}}} - R_u T_{\text{prod}}) + c(\bar{h}_{\text{O}_2@T_{\text{prod}}} - R_u T_{\text{prod}}) \end{aligned}$$

The relation for the final pressure is

$$P_2 = \frac{N_{\text{tot}}}{N_1} \frac{T_{\text{prod}}}{T_{\text{reac}}} P_1 = \left(\frac{a+b+c}{2} \right) \left(\frac{T_{\text{prod}}}{298.15 \text{ K}} \right) (101.3 \text{ kPa})$$

Solving all the equations simultaneously using EES, we obtain the final temperature and pressure in the tank to be

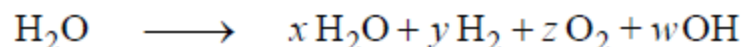
$$\begin{aligned} T_{\text{prod}} &= \mathbf{3857 \text{ K}} \\ P_2 &= \mathbf{1043 \text{ kPa}} \end{aligned}$$

Simultaneous Reactions

16-40 One mole of H_2O is heated to 3400 K at a pressure of 1 atm. Determine the equilibrium composition, assuming that only H_2O , OH , O_2 , and H_2 are present. *Answers:* $0.574\text{H}_2\text{O}$, 0.308H_2 , 0.095O_2 , 0.236OH

Assumptions 1 The equilibrium composition consists of H_2O , OH , O_2 , and H_2 . 2 The constituents of the mixture are ideal gases.

Analysis The reaction equation during this process can be expressed as



Mass balances for hydrogen and oxygen yield

$$\text{H balance:} \quad 2 = 2x + 2y + w \quad (1)$$

$$\text{O balance:} \quad 1 = x + 2z + w \quad (2)$$

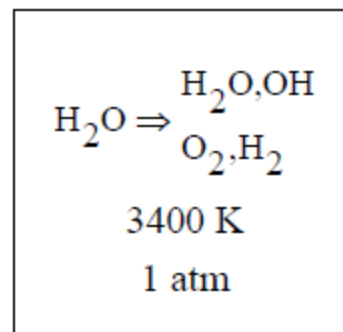
The mass balances provide us with only two equations with four unknowns, and thus we need to have two more equations (to be obtained from the K_p relations) to determine the equilibrium composition of the mixture. They are



The equilibrium constant for these two reactions at 3400 K are determined from Table A-28 to be

$$\ln K_{p1} = -1.891 \longrightarrow K_{p1} = 0.15092$$

$$\ln K_{p2} = -1.576 \longrightarrow K_{p2} = 0.20680$$



The K_p relations for these two simultaneous reactions are

$$K_{P1} = \frac{N_{\text{H}_2}^{\nu_{\text{H}_2}} N_{\text{O}_2}^{\nu_{\text{O}_2}}}{N_{\text{H}_2\text{O}}^{\nu_{\text{H}_2\text{O}}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{\text{H}_2} + \nu_{\text{O}_2} - \nu_{\text{H}_2\text{O}})} \quad \text{and} \quad K_{P2} = \frac{N_{\text{H}_2}^{\nu_{\text{H}_2}} N_{\text{OH}}^{\nu_{\text{OH}}}}{N_{\text{H}_2\text{O}}^{\nu_{\text{H}_2\text{O}}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{\text{H}_2} + \nu_{\text{OH}} - \nu_{\text{H}_2\text{O}})}$$

where $N_{\text{total}} = N_{\text{H}_2\text{O}} + N_{\text{H}_2} + N_{\text{O}_2} + N_{\text{OH}} = x + y + z + w$

Substituting,

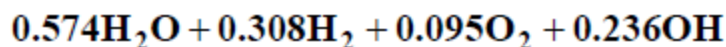
$$0.15092 = \frac{(y)(z)^{1/2}}{x} \left(\frac{1}{x + y + z + w} \right)^{1/2} \quad (3)$$

$$0.20680 = \frac{(w)(y)^{1/2}}{x} \left(\frac{1}{x + y + z + w} \right)^{1/2} \quad (4)$$

Solving Eqs. (1), (2), (3), and (4) simultaneously for the four unknowns x , y , z , and w yields

$$x = 0.574 \quad y = 0.308 \quad z = 0.095 \quad w = 0.236$$

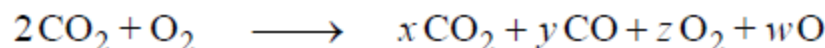
Therefore, the equilibrium composition becomes



16–41 A mixture of 2 mol of CO_2 and 1 mol of O_2 is heated to 3200 K at a pressure of 2 atm. Determine the equilibrium composition of the mixture, assuming that only CO_2 , CO , O_2 , and O are present.

Assumptions 1 The equilibrium composition consists of CO_2 , CO , O_2 , and O . 2 The constituents of the mixture are ideal gases.

Analysis The reaction equation during this process can be expressed as



$\text{CO}_2, \text{CO}, \text{O}_2, \text{O}$ 3200 K 2 atm

Mass balances for carbon and oxygen yield

C balance: $2 = x + y$ (1)

O balance: $6 = 2x + y + 2z + w$ (2)

The mass balances provide us with only two equations with four unknowns, and thus we need to have two more equations (to be obtained from the K_P relations) to determine the equilibrium composition of the mixture. They are



The equilibrium constant for these two reactions at 3200 K are determined from Table A-28 to be

$$\ln K_{P1} = -0.429 \longrightarrow K_{P1} = 0.65116$$

$$\ln K_{P2} = -3.072 \longrightarrow K_{P2} = 0.04633$$



The K_P relations for these two simultaneous reactions are

$$K_{P1} = \frac{N_{\text{CO}}^{\nu_{\text{CO}}} N_{\text{O}_2}^{\nu_{\text{O}_2}}}{N_{\text{CO}_2}^{\nu_{\text{CO}_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{\text{CO}} + \nu_{\text{O}_2} - \nu_{\text{CO}_2})}$$

$$K_{P2} = \frac{N_{\text{O}}^{\nu_{\text{O}}}}{N_{\text{O}_2}^{\nu_{\text{O}_2}}} \left(\frac{P}{N_{\text{total}}} \right)^{\nu_{\text{O}} - \nu_{\text{O}_2}}$$

where

$$N_{\text{total}} = N_{\text{CO}_2} + N_{\text{O}_2} + N_{\text{CO}} + N_{\text{O}} = x + y + z + w$$

Substituting,

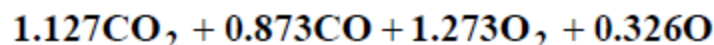
$$0.65116 = \frac{(y)(z)^{1/2}}{x} \left(\frac{2}{x + y + z + w} \right)^{1/2} \quad (3)$$

$$0.04633 = \frac{w^2}{z} \left(\frac{2}{x + y + z + w} \right)^{2-1} \quad (4)$$

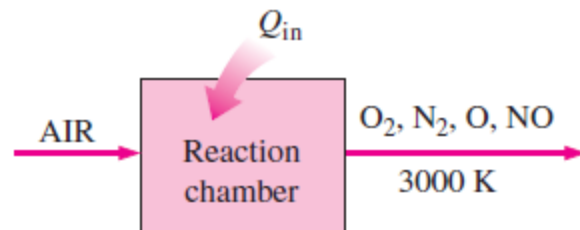
Solving Eqs. (1), (2), (3), and (4) simultaneously for the four unknowns x , y , z , and w yields

$$x = 1.127 \quad y = 0.873 \quad z = 1.273 \quad w = 0.326$$

Thus the equilibrium composition is



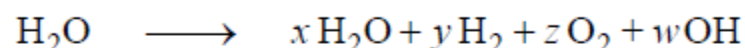
16–42 Air (21 percent O_2 , 79 percent N_2) is heated to 3000 K at a pressure of 2 atm. Determine the equilibrium composition, assuming that only O_2 , N_2 , O , and NO are present. Is it realistic to assume that no N will be present in the final equilibrium mixture?



16–45 Water vapor (H_2O) is heated during a steady-flow process at 1 atm from 298 to 3000 K at a rate of 0.2 kg/min. Determine the rate of heat supply needed during this process, assuming (a) some H_2O dissociates into H_2 , O_2 , and OH and (b) no dissociation takes place. *Answers: (a) 2055 kJ/min, (b) 1404 kJ/min*

Assumptions 1 The equilibrium composition consists of H_2O , OH , O_2 , and H_2 . 2 The constituents of the mixture are ideal gases.

Analysis (a) Assuming some H_2O dissociates into H_2 , O_2 , and O , the dissociation equation can be written as



Mass balances for hydrogen and oxygen yield

$$\text{H balance:} \quad 2 = 2x + 2y + w \quad (1)$$

$$\text{O balance:} \quad 1 = x + 2z + w \quad (2)$$

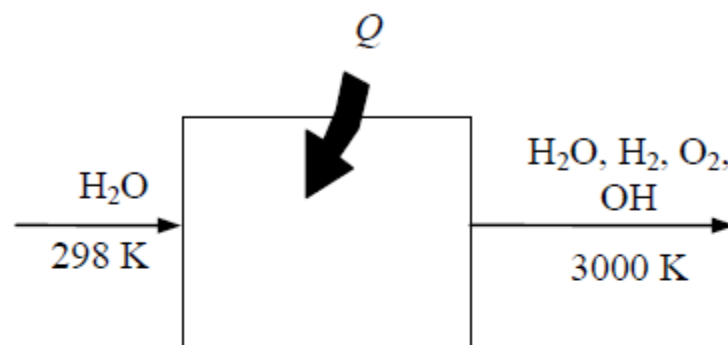
The mass balances provide us with only two equations with four unknowns, and thus we need to have two more equations (to be obtained from the K_P relations) to determine the equilibrium composition of the mixture. They are



The equilibrium constant for these two reactions at 3000 K are determined from Table A-28 to be

$$\ln K_{P1} = -3.086 \longrightarrow K_{P1} = 0.04568$$

$$\ln K_{P2} = -2.937 \longrightarrow K_{P2} = 0.05302$$



The K_P relations for these three simultaneous reactions are

$$K_{P1} = \frac{N_{\text{H}_2}^{\nu_{\text{H}_2}} N_{\text{O}_2}^{\nu_{\text{O}_2}}}{N_{\text{H}_2\text{O}}^{\nu_{\text{H}_2\text{O}}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{\text{H}_2} + \nu_{\text{O}_2} - \nu_{\text{H}_2\text{O}})}$$

$$K_{P2} = \frac{N_{\text{H}_2}^{\nu_{\text{H}_2}} N_{\text{OH}}^{\nu_{\text{OH}}}}{N_{\text{H}_2\text{O}}^{\nu_{\text{H}_2\text{O}}}} \left(\frac{P}{N_{\text{total}}} \right)^{(\nu_{\text{H}_2} + \nu_{\text{O}_2} - \nu_{\text{H}_2\text{O}})}$$

where

$$N_{\text{total}} = N_{\text{H}_2\text{O}} + N_{\text{H}_2} + N_{\text{O}_2} + N_{\text{OH}} = x + y + z + w$$

Substituting,

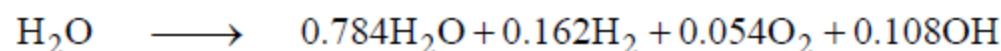
$$0.04568 = \frac{(y)(z)^{1/2}}{x} \left(\frac{1}{x + y + z + w} \right)^{1/2} \quad (3)$$

$$0.05302 = \frac{(w)(y)^{1/2}}{x} \left(\frac{1}{x + y + z + w} \right)^{1/2} \quad (4)$$

Solving Eqs. (1), (2), (3), and (4) simultaneously for the four unknowns x , y , z , and w yields

$$x = 0.784 \quad y = 0.162 \quad z = 0.054 \quad w = 0.108$$

Thus the balanced equation for the dissociation reaction is



The heat transfer for this dissociation process is determined from the steady-flow energy balance

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}} \text{ with } W = 0,$$

$$Q_{\text{in}} = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R$$

Assuming the O₂ and O to be ideal gases, we have $h = h(T)$. From the tables,

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{298\text{ K}}$ kJ/kmol	$\bar{h}_{3000\text{ K}}$ kJ/kmol
H ₂ O	-241,820	9904	136,264
H ₂	0	8468	97,211
O ₂	0	8682	106,780
OH	39,460	9188	98,763

Substituting,

$$\begin{aligned} Q_{\text{in}} &= 0.784(-241,820 + 136,264 - 9904) \\ &\quad + 0.162(0 + 97,211 - 8468) \\ &\quad + 0.054(0 + 106,780 - 8682) \\ &\quad + 0.108(39,460 + 98,763 - 9188) - (-241,820) \\ &= 184,909 \text{ kJ / kmol H}_2\text{O} \end{aligned}$$

The mass flow rate of H₂O can be expressed in terms of the mole numbers as

$$\dot{N} = \frac{\dot{m}}{M} = \frac{0.2 \text{ kg / min}}{18 \text{ kg / kmol}} = 0.01111 \text{ kmol / min}$$

Thus,

$$\dot{Q}_{\text{in}} = \dot{N} \times Q_{\text{in}} = (0.01111 \text{ kmol/min})(184,909 \text{ kJ/kmol}) = \mathbf{2055 \text{ kJ/min}}$$

(b) If no dissociates takes place, then the process involves no chemical reactions and the heat transfer can be determined from the steady-flow energy balance for nonreacting systems to be

$$\begin{aligned} \dot{Q}_{\text{in}} &= \dot{m}(h_2 - h_1) = \dot{N}(\bar{h}_2 - \bar{h}_1) \\ &= (0.01111 \text{ kmol / min})(136,264 - 9904) \text{ kJ / kmol} \\ &= \mathbf{1404 \text{ kJ / min}} \end{aligned}$$

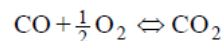
16-47

Ethyl alcohol ($\text{C}_2\text{H}_5\text{OH}(g)$) at 25°C is burned in a steady-flow adiabatic combustion chamber with 40 percent excess air that also enters at 25°C . Determine the adiabatic flame temperature of the products at 1 atm assuming the significant equilibrium reactions are $\text{CO}_2 = \text{CO} + \frac{1}{2}\text{O}_2$ and $\frac{1}{2}\text{N}_2 + \frac{1}{2}\text{O}_2 = \text{NO}$. Plot the adiabatic flame temperature and kmols of CO_2 , CO , and NO at equilibrium for values of percent excess air between 10 and 100 percent.

Variations of K_p with Temperature

16–50 Estimate the enthalpy of reaction $\bar{h}_R$ for the combustion process of carbon monoxide at 2200 K, using (a) enthalpy data and (b) K_p data.

Analysis (a) The complete combustion equation of CO can be expressed as



The $\bar{h}_R$ of the combustion process of CO at 2200 K is the amount of energy released as one kmol of CO is burned in a steady-flow combustion chamber at a temperature of 2200 K, and can be determined from

$$\bar{h}_R = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R$$

Assuming the CO, O₂ and CO₂ to be ideal gases, we have $h = h(T)$. From the tables,

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{298 \text{ K}}$ kJ/kmol	$\bar{h}_{2200 \text{ K}}$ kJ/kmol
CO ₂	-393,520	9364	112,939
CO	-110,530	8669	72,688
O ₂	0	8682	75,484

Substituting,

$$\begin{aligned} \bar{h}_R &= 1(-393,520 + 112,939 - 9364) \\ &\quad - 1(-110,530 + 72,688 - 8669) \\ &\quad - 0.5(0 + 75,484 - 8682) \\ &= -276,835 \text{ kJ / kmol} \end{aligned}$$

(b) The $\bar{h}_R$ value at 2200 K can be estimated by using K_p values at 2000 K and 2400 K (the closest two temperatures to 2200 K for which K_p data are available) from Table A-28,

$$\ln \frac{K_{p2}}{K_{p1}} \cong \frac{\bar{h}_R}{R_u} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \quad \text{or} \quad \ln K_{p2} - \ln K_{p1} \cong \frac{\bar{h}_R}{R_u} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$3.860 - 6.635 \cong \frac{\bar{h}_R}{8.314 \text{ kJ/kmol} \cdot \text{K}} \left(\frac{1}{2000 \text{ K}} - \frac{1}{2400 \text{ K}} \right)$$

$$\bar{h}_R \cong -276,856 \text{ kJ/kmol}$$

16-52 Using the enthalpy of reaction $\bar{h}_R$ data and the K_p value at 2400 K, estimate the K_p value of the combustion process $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{H}_2\text{O}$ at 2600 K. *Answer: 104.1*

16-53 Estimate the enthalpy of reaction $\bar{h}_R$ for the dissociation process $\text{CO}_2 \rightleftharpoons \text{CO} + \frac{1}{2}\text{O}_2$ at 2200 K, using (a) enthalpy data and (b) K_p data.

16-54 Estimate the enthalpy of reaction $\bar{h}_R$ for the dissociation process $\text{O}_2 \rightleftharpoons 2\text{O}$ at 3100 K, using (a) enthalpy data and (b) K_p data. *Answers: (a) 513,614 kJ/kmol, (b) 512,808 kJ/kmol*

16-55 Estimate the enthalpy of reaction for the equilibrium reaction $\text{CH}_4 + 2\text{O}_2 = \text{CO}_2 + 2\text{H}_2\text{O}$ at 2500 K, using (a) enthalpy data and (b) K_p data.

Phase Equilibrium

16-61 Show that a mixture of saturated liquid water and saturated water vapor at 100°C satisfies the criterion for phase equilibrium.

Analysis Using the definition of Gibbs function and enthalpy and entropy data from Table A-4,

$$g_f = h_f - Ts_f = (419.17 \text{ kJ/kg}) - (373.15 \text{ K})(1.3072 \text{ kJ/kg} \cdot \text{K}) = -68.61 \text{ kJ/kg}$$

$$g_g = h_g - Ts_g = (2675.6 \text{ kJ/kg}) - (373.15 \text{ K})(7.3542 \text{ kJ/kg} \cdot \text{K}) = -68.62 \text{ kJ/kg}$$

which are practically same. Therefore, the criterion for phase equilibrium is satisfied.

16-62 Show that a mixture of saturated liquid water and saturated water vapor at 300 kPa satisfies the criterion for phase equilibrium.

Analysis The saturation temperature at 300 kPa is 406.7 K. Using the definition of Gibbs function and enthalpy and entropy data from Table A-5,

$$g_f = h_f - Ts_f = (561.43 \text{ kJ/kg}) - (406.7 \text{ K})(1.6717 \text{ kJ/kg} \cdot \text{K}) = -118.5 \text{ kJ/kg}$$

$$g_g = h_g - Ts_g = (2724.9 \text{ kJ/kg}) - (406.7 \text{ K})(6.9917 \text{ kJ/kg} \cdot \text{K}) = -118.6 \text{ kJ/kg}$$

which are practically same. Therefore, the criterion for phase equilibrium is satisfied.

16-63 Show that a saturated liquid–vapor mixture of refrigerant-134a at -10°C satisfies the criterion for phase equilibrium.

16-64 Consider a mixture of oxygen and nitrogen in the gas phase. How many independent properties are needed to fix the state of the system? *Answer: 3*

16–66 Consider a liquid–vapor mixture of ammonia and water in equilibrium at 25°C. If the composition of the liquid phase is 50 percent NH₃ and 50 percent H₂O by mole numbers, determine the composition of the vapor phase of this mixture. Saturation pressure of NH₃ at 25°C is 1003.5 kPa.

Answers: 0.31 percent, 99.69 percent

Assumptions The mixture is ideal and thus Raoult's law is applicable.

Properties At 25°C, $P_{\text{sat,H}_2\text{O}} = 3.170 \text{ kPa}$ and $P_{\text{sat,NH}_3} = 1003.5 \text{ kPa}$.

Analysis The vapor pressures are

$$P_{\text{H}_2\text{O}} = y_{f,\text{H}_2\text{O}} P_{\text{sat,H}_2\text{O}}(T) = 0.50(3.170 \text{ kPa}) = 1.585 \text{ kPa}$$

$$P_{\text{NH}_3} = y_{f,\text{NH}_3} P_{\text{sat,NH}_3}(T) = 0.50(1003.5 \text{ kPa}) = 501.74 \text{ kPa}$$

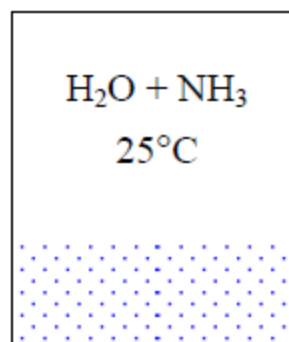
Thus the total pressure of the mixture is

$$P_{\text{total}} = P_{\text{H}_2\text{O}} + P_{\text{NH}_3} = (1.585 + 501.74) \text{ kPa} = 503.33 \text{ kPa}$$

Then the mole fractions in the vapor phase become

$$y_{g,\text{H}_2\text{O}} = \frac{P_{\text{H}_2\text{O}}}{P_{\text{total}}} = \frac{1.585 \text{ kPa}}{503.33 \text{ kPa}} = \mathbf{0.0031} \text{ or } 0.31\%$$

$$y_{g,\text{NH}_3} = \frac{P_{\text{NH}_3}}{P_{\text{total}}} = \frac{501.74 \text{ kPa}}{503.33 \text{ kPa}} = \mathbf{0.9969} \text{ or } 99.69\%$$



16-67 A two-phase mixture of ammonia and water is in equilibrium at 50°C . If the composition of the vapor phase is 99 percent NH_3 and 1 percent H_2O by mole numbers, determine the composition of the liquid phase of this mixture. Saturation pressure of NH_3 at 50°C is 2033.5 kPa.

16-68 Using the liquid–vapor equilibrium diagram of an oxygen–nitrogen mixture, determine the composition of each phase at 80 K and 100 kPa.

16-73 A wall made of natural rubber separates O_2 and N_2 gases at 25°C and 500 kPa. Determine the molar concentrations of O_2 and N_2 in the wall.

16-74 Consider a glass of water in a room at 27°C and 97 kPa. If the relative humidity in the room is 100 percent and the water and the air are in thermal and phase equilibrium, determine (a) the mole fraction of the water vapor in the air and (b) the mole fraction of air in the water.

Assumptions 1 Both the air and water vapor are ideal gases. 2 Air is saturated since the humidity is 100 percent. 3 Air is weakly soluble in water and thus Henry's law is applicable.

Properties The saturation pressure of water at 27°C is 3.568 kPa (Table A-4). Henry's constant for air dissolved in water at 27°C (300 K) is given in Table 16-2 to be $H = 74,000$ bar. Molar masses of dry air and water are 29 and 18 kg/kmol, respectively (Table A-1).

Analysis (a) Noting that air is saturated, the partial pressure of water vapor in the air will simply be the saturation pressure of water at 27°C,

$$P_{\text{vapor}} = P_{\text{sat}@27^\circ\text{C}} = 3.600 \text{ kPa} \quad (\text{Table A-4})$$

Assuming both the air and vapor to be ideal gases, the mole fraction of water vapor in the air is determined to be

$$y_{\text{vapor}} = \frac{P_{\text{vapor}}}{P} = \frac{3.600 \text{ kPa}}{97 \text{ kPa}} = 0.0371$$

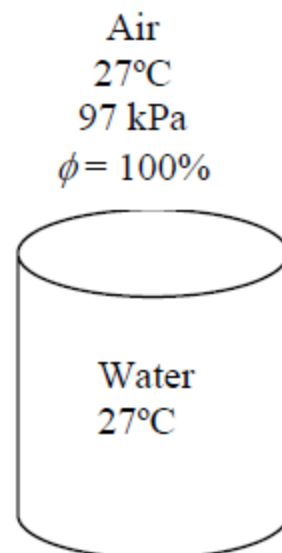
(b) Noting that the total pressure is 97 kPa, the partial pressure of dry air is

$$P_{\text{dry air}} = P - P_{\text{vapor}} = 97 - 3.600 = 93.4 \text{ kPa} = 0.934 \text{ bar}$$

From Henry's law, the mole fraction of air in the water is determined to be

$$y_{\text{dry air, liquid side}} = \frac{P_{\text{dry air, gas side}}}{H} = \frac{0.934 \text{ bar}}{74,000 \text{ bar}} = 1.26 \times 10^{-5}$$

Discussion The amount of air dissolved in water is very small, as expected.



16–76 Consider a carbonated drink in a bottle at 27°C and 130 kPa. Assuming the gas space above the liquid consists of a saturated mixture of CO_2 and water vapor and treating the drink as water, determine (a) the mole fraction of the water vapor in the CO_2 gas and (b) the mass of dissolved CO_2 in a 300-ml drink.

Chapter 16

End of the Problems